Capturing 98% Hubbard Mottness with a Momentum-Mixing Hatsugai–Kohmoto Model

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(Dated: May 4, 2026)

We benchmark a momentum-mixing Hatsugai–Kohmoto construction against established zero-temperature results for the two-dimensional square-lattice Hubbard model and use the same calculation to diagnose the Luttinger surface. At half filling and $t' = 0$, the calculated ground-state energy, double occupancy, and antiferromagnetic moment reproduce the interaction dependence of AFQMC and DMET reference data over $U/t = 2$–12. The agreement is especially close for the local double occupancy at intermediate and strong coupling, while the magnetic moment retains the larger method and finite-size sensitivity familiar from Hubbard benchmarks. Away from half filling, and for finite next-nearest-neighbor hopping $t' = -0.25t$, the model captures the suppression of double occupancy, the persistence of antiferromagnetic correlations, and the evolution of the chemical potential across $n = 0.8$–1.0. We then examine the zero-frequency Green's function and identify a surface of zeros that reconstructs the low-energy sign structure of $G(\mathbf{k}, \omega = 0)$. These results connect conventional static benchmarks to a momentum-space diagnostic of Mottness in a strongly correlated lattice model.

I. INTRODUCTION

The two-dimensional Hubbard model remains a central testbed for strongly correlated electron physics because its minimal ingredients already contain the competing tendencies of itinerancy, magnetism, Mottness, and unconventional low-energy structure[1,2]. It is also a demanding benchmark problem. Ground-state energies can agree across high-accuracy methods while order parameters, correlation functions, and momentum-resolved quantities remain more sensitive to finite-size effects, constraints, and the representation of correlations[3]. A useful approximate construction should therefore be judged not only by whether it reproduces a single number, but by whether it gives a coherent account of local observables, magnetic correlations, and the momentum-space structure of the one-particle Green's function.

The latter point is particularly important in a Mott regime. In a weakly interacting Fermi liquid, the sign change of the zero-frequency Green's function is tied to a Fermi surface of poles. In a Mott insulator, by contrast, the low-energy sign structure can be controlled by zeros of G rather than poles[4,5]. The corresponding Luttinger surface is a compact diagnostic of Mottness: it records where the Green's function changes sign even though the spectral weight at the chemical potential has been removed. Demonstrating this structure in an approximate model requires a baseline check that the same model also reproduces conventional Hubbard observables.

Here we take this route for a momentum-mixing extension of the exactly solvable Hatsugai–Kohmoto construction[6]. We first compare the MMHK results with established half-filled Hubbard benchmarks at $t' = 0$, using the ground-state energy, double occupancy, and antiferromagnetic moment. We then examine doped cases and finite next-nearest-neighbor hopping $t' = -0.25t$, where frustration and density tuning modify the magnetic response. Finally, we analyze $\mathrm{Re}\, G(\mathbf{k}, 0)$ and extract the Luttinger surface from its sign changes. This sequence provides a controlled bridge from static benchmark observables to the momentum-space topology of Green's-function zeros.

II. MODEL AND METHOD

We consider the square-lattice Hubbard Hamiltonian

$$\begin{aligned} H = & -t \sum_{\langle ij \rangle, \sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + \mathrm{H.c.} \right) \\ & - t' \sum_{\langle\langle ij \rangle\rangle, \sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + \mathrm{H.c.} \right) \\ & + U \sum_i n_{i\uparrow} n_{i\downarrow} - \mu \sum_{i,\sigma} n_{i\sigma}. \end{aligned} \tag{1}$$

The benchmark calculations use 4×4 clusters and scan total momentum sectors to identify the ground state in each particle-number sector. For fixed target density, the chemical potential is determined by solving the grand-canonical density condition; at half filling particle-hole symmetry fixes $\mu = U/2$ for $t' = 0$. The MMHK construction keeps the interaction local in the momentum-space cluster basis while allowing the mixing needed to represent Hubbard-model correlations beyond the purely diagonal Hatsugai–Kohmoto limit.

The observables reported below are the energy per site, the local double occupancy

$$D = \frac{1}{N} \sum_i \langle n_{i\uparrow} n_{i\downarrow} \rangle, \tag{2}$$

and the magnetic structure-factor-derived moment used in the benchmark tables. Momentum-resolved Green's

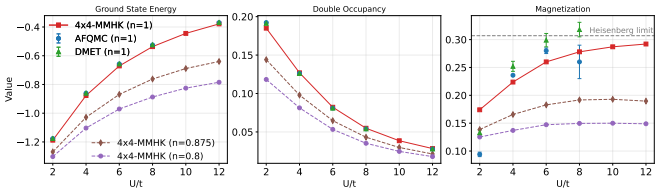


FIG. 1. Half-filled benchmark at $t' = 0$. Ground-state energy, double occupancy, and magnetic moment are compared with AFQMC and DMET reference values from Ref. 3. The MMHK calculation captures the interaction dependence of all three observables; local double occupancy is especially close to the benchmark data for $U/t \geq 4$.

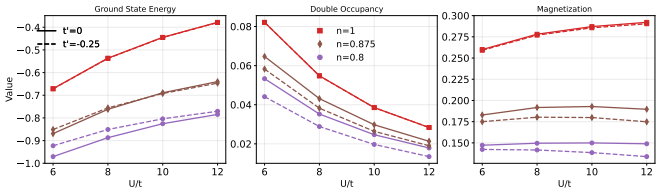


FIG. 2. Doped and frustrated cases. Comparison of $t' = 0$ and $t' = -0.25t$ for target densities $n = 0.8$, 0.875, and 1.0. The finite next-nearest-neighbor hopping suppresses magnetic correlations away from half filling while preserving the strong-coupling reduction of double occupancy.

functions are evaluated from particle-addition and particle-removal matrix elements generated in the neighboring particle-number sectors. The zero-frequency Green's function is then used to define the sign field

$$s(\mathbf{k}) = \mathrm{sgn}\,\mathrm{Re}\,G(\mathbf{k}, 0), \tag{3}$$

and the Luttinger surface is identified by sign changes of $s(\mathbf{k})$ not associated with a quasiparticle pole.

III. HALF-FILLED BENCHMARK

Figure 1 compares the MMHK half-filled results with AFQMC and DMET benchmarks. The energy follows the same monotonic increase toward the atomic limit as the reference calculations. The MMHK energy is systematically lower than the AFQMC and DMET thermodynamic-limit estimates by roughly 0.009–0.017t over the range studied, corresponding to relative deviations of about 1–3%. This offset is largest in relative terms at stronger coupling, but remains smooth and does not obscure the trend with U.

The local double occupancy is more tightly reproduced. For $U/t \geq 4$, the absolute deviation from AFQMC is between 5.7×10^{-4} and 1.1×10^{-3}, or roughly 0.6–2.0%. The largest discrepancy occurs at $U/t = 2$, where MMHK gives $D = 0.1850$ compared with the AFQMC value 0.1923(3). The magnetic moment shows the expected growth with interaction strength, from $m = 0.174$ at $U/t = 2$ to $m = 0.292$ at $U/t = 12$. As in previous benchmark studies, magnetization is a more sensitive quantity than the energy or double occupancy, and discrepancies should be interpreted in light of finite-size and order-parameter sensitivity.

IV. DOPING AND NEXT-NEAREST-NEIGHBOR HOPPING

The benchmark can be extended away from half filling by tuning μ to target densities $n = 0.8$ and $n = 0.875$. At $t' = 0$, increasing U suppresses double occupancy and drives the magnetic moment toward a broad plateau. For example, at $n = 0.8$ the moment changes only weakly from $m = 0.147$ at $U/t = 6$ to $m = 0.149$ at $U/t = 12$, while D decreases from 0.0533 to 0.0179. At $n = 0.875$, the moment is larger, reaching $m \simeq 0.19$ for $U/t = 8$–12, consistent with stronger magnetic correlations closer to half filling.

Finite $t' = -0.25t$ modifies this evolution. At the same density and interaction, the magnetic moment is generally reduced relative to $t' = 0$, most clearly at strong coupling and lower density. At $n = 0.8$ and $U/t = 12$, for instance, the moment decreases from 0.149 at $t' = 0$ to 0.134 at $t' = -0.25t$. The local double occupancy is also reduced, from 0.0179 to 0.0135, reflecting the combined effect of interaction strength and modified band structure.

V. LUTTINGER SURFACE

Having established the behavior of static observables, we turn to the zero-frequency Green's function. The key quantity is the sign of $\mathrm{Re}\,G(\mathbf{k}, 0)$ over the Brillouin zone. A sign change can occur through a pole, as for a Fermi surface, or through a zero, as for a Luttinger surface. In the correlated insulating regime the latter mechanism is expected to dominate because the self-energy diverges at the zero surface.

In the present calculation, the Luttinger surface is obtained by evaluating $G(\mathbf{k}, \omega = 0)$ on the cluster momentum grid and, where needed, interpolating the sign field between neighboring momenta. The benchmarked half-filled calculation provides the cleanest setting: the zero surface appears near the noninteracting Fermi-surface contour expected from particle-hole symmetry, while the spectral weight at zero energy is removed by the interaction-generated gap. This separation between the sign structure of G and the absence of a pole is the defining signature of the Luttinger surface.

Figure 3 shows representative spectral and Luttinger-surface diagnostics for $t' = 0$. Increasing U/t removes low-energy spectral weight while leaving a well-defined zero contour in the zero-frequency propagator. At large U/t, the fitted Luttinger-surface dispersion approaches the noninteracting form, consistent with a Mott state in

which the sign structure remains tied to the underlying band topology even though the quasiparticle pole is absent.

For $t' = -0.25t$, the zero surface is no longer constrained to the particle-hole-symmetric contour. Figure 4 shows that the contour shifts with the modified band structure, while the same zero-of-G criterion remains applicable. The resulting sign map provides a compact way to connect the static benchmark observables to momentum-space Mottness: double occupancy and magnetization diagnose the local and spin response, while the Luttinger surface diagnoses the topology of the zero-frequency propagator.

VI. DISCUSSION

The benchmark results show that the MMHK construction captures the main zero-temperature Hubbard trends across weak, intermediate, and strong coupling. The most robust agreement is found for local quantities such as double occupancy, while the energy contains a smooth systematic offset and the magnetic moment remains the most sensitive observable. This hierarchy is useful rather than merely diagnostic: it clarifies which conclusions can be drawn directly from the benchmark and which require finite-size or method-comparison checks.

The Luttinger-surface result is the natural extension of this benchmark. A method may reproduce local observables yet still give an incorrect momentum-space sign structure. By computing $G(\mathbf{k}, \omega = 0)$ directly after validating the static quantities, we can test whether the correlated state has the expected zero surface and how that surface evolves under doping and band-structure frustration. In this sense, the Luttinger surface is not an isolated observation; it is the momentum-resolved counterpart of the benchmarked Mott physics.

VII. CONCLUSION

We have benchmarked a momentum-space Hubbard-cluster calculation against established half-filled Hubbard results and extended it to doped and frustrated regimes. The method reproduces the interaction dependence of energy, double occupancy, and magnetization, with particularly good agreement for double occupancy at intermediate and strong coupling. Using the same calculation, we identify the Luttinger surface from the sign changes of the zero-frequency Green's function. This connects conventional static benchmarks to a momentum-space diagnostic of Mottness and provides a practical route for studying Green's-function zeros in correlated lattice models.

ACKNOWLEDGMENTS

We thank [names] for useful discussions. This work used computational resources at [facility].

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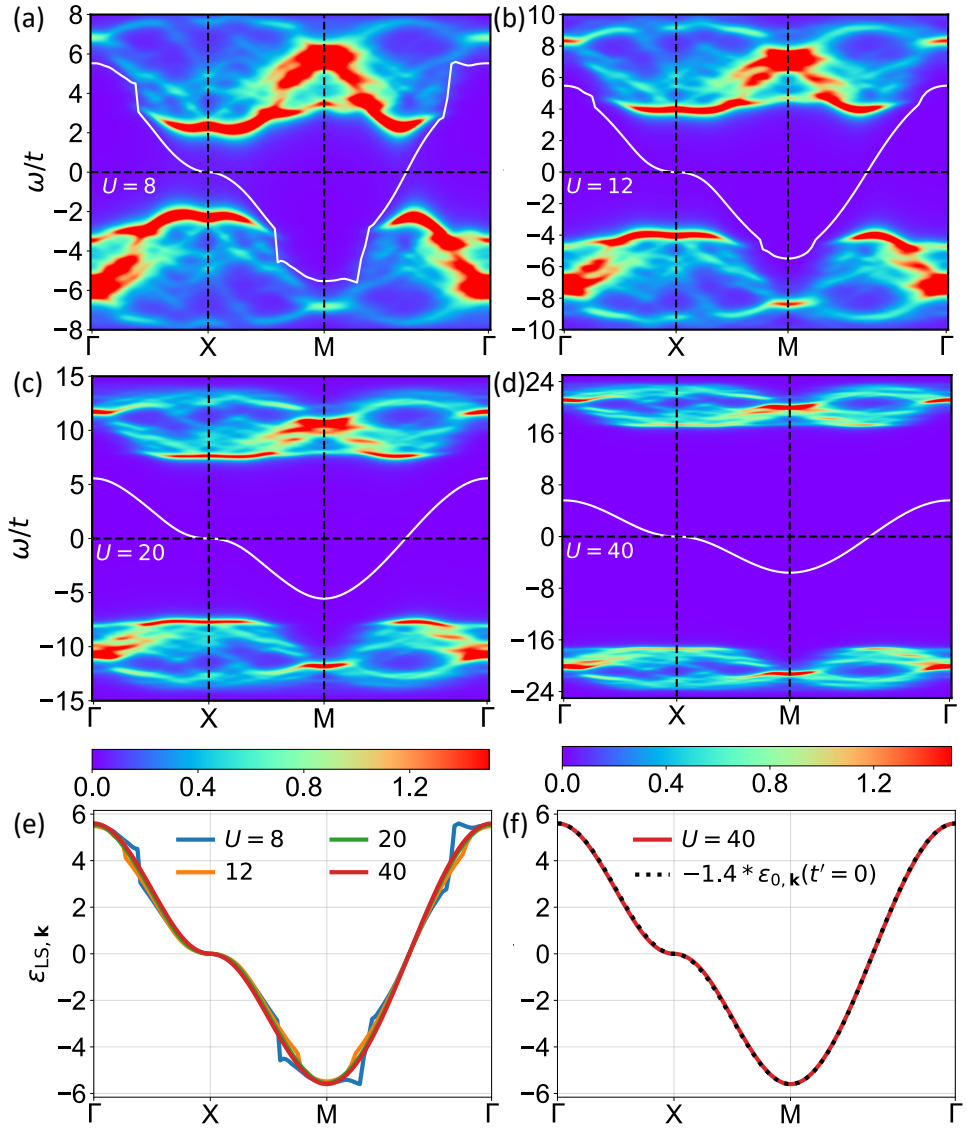


FIG. 3. Spectral function and Luttinger surface for $t' = 0$ at several interaction strengths. The zero contour of $G(\mathbf{k}, 0)$ survives as the low-energy spectral weight is suppressed, distinguishing the Luttinger surface from a Fermi surface of poles.

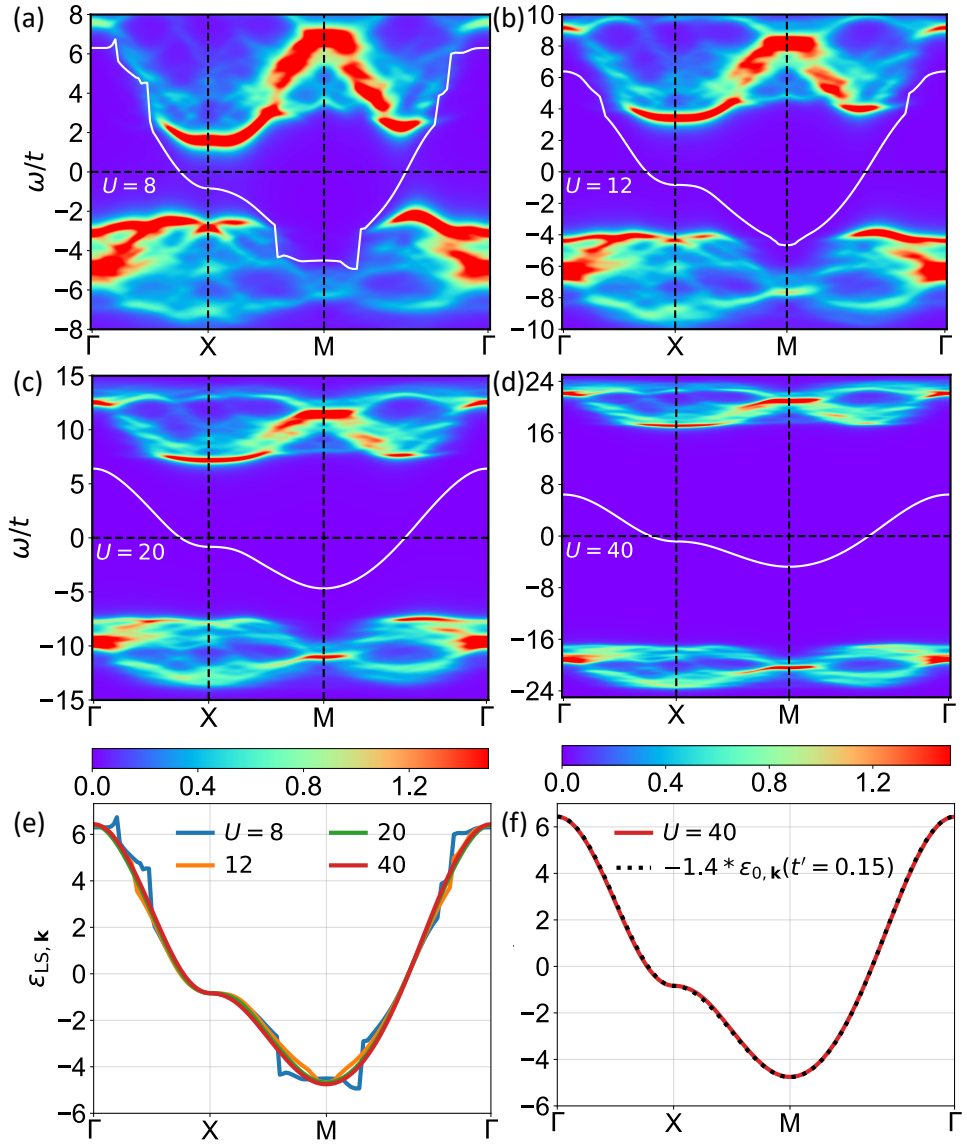


FIG. 4. Spectral function and Luttinger surface for $t' = -0.25t$. Next-nearest-neighbor hopping shifts the zero contour away from the particle-hole-symmetric reference while preserving the interpretation of the sign change as a zero of the Green's function rather than a quasiparticle pole.